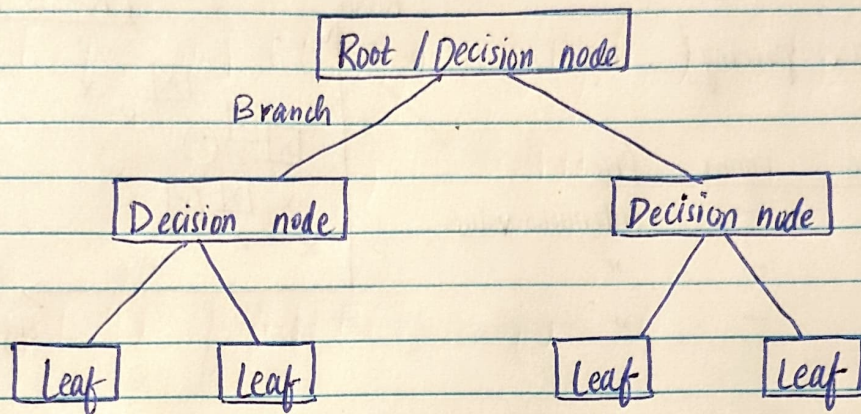


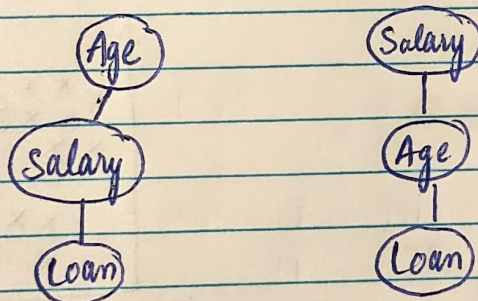
5, Decision Tree - { classification, Regression }



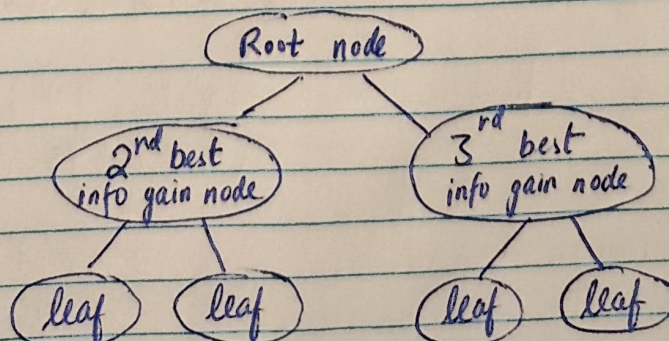
- You need to decide which root node to be selected from the decision node

Age	Salary	Loan
23	12,000	No
40	1,65,000	Yes

- Here what will be our root node is it age or salary



- In order to tackle this we find the entropy of the entire dataset, then we need to calculate the information gain of each node and whoever has the highest info gain will be selected as root node.
- We use entropy / information gain to select the root node. mostly we use entropy



Information gain is a measure of how much a feature provides info about a class.

It calculates difference between Entropy before split and average entropy after split.

$$E(S) \{ \text{Entropy} \} = \sum_{i=1}^C -P_i \log_2(P_i)$$

$$\{ \text{Gini impurity} \} G(S) = 1 - \sum_{i=1}^C (P_i)^2$$

$$\{ \text{Information Gain} \} IG(S, F) = \{ \text{Entropy} \} E(S) - \sum_{y \in F} \frac{|S_y|}{|S|} E(S_y)$$

Example:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	overcast	Mild	High	Strong	Yes
D13	overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

(i) Entropy (Dataset)

$$S = [9 \text{ Yes}, 5 \text{ No}] \cdot E(S) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right)$$

$$E(S) = 0.94$$

② Outlook : Sunny / overcast / Rain

$$S_{\text{Sunny}} [2 \text{ Yes}, 3 \text{ No}] E(S_{\text{Sunny}}) = -\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right)$$

$$E(S_{\text{Sunny}}) = 0.971$$

$$S_{\text{overcast}} [4 \text{ Yes}, 0 \text{ No}] E(S_{\text{overcast}}) = 0$$

$$S_{\text{Rain}} [3 \text{ Yes}, 2 \text{ No}] E(S_{\text{Rain}}) = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right)$$

$$E(S_{\text{Rain}}) = 0.971$$

$$\textcircled{3} \text{ Gain}(S, \text{outlook}) = E(S) - \sum_{s \in \{\text{Sunny, overcast, rain}\}} \frac{|S_s|}{|S|} E(S_s)$$

$$= 0.94 - \frac{5}{14} E_{\text{Sunny}} - \frac{4}{14} E_{\text{overcast}} - \frac{5}{14} E_{\text{Rain}}$$

$$= 0.94 - \frac{5}{14} (0.971) - \frac{4}{14} (0) - \frac{5}{14} (0.971)$$

$$G(S, \text{outlook}) = 0.24$$

Similarly if we calculate the gain for temperature, humidity and wind, which ever value has gain higher that will be selected as Root Node.

Interview
question

- 1) What is .fit()
- 2) What is transform()
- 3) What is fit_transform

Students = ['Ram', 'Sam', 'Tom']

i.e. fit(students)

↓

Ram → 0

Sam → 1

Tom → 2

The encoder looks at the data and memorizes it. It does not return output, only learns the mapping

• `transform()` ↘

now the encoder already knows

Ram → 0

Sam → 1

Tom → 2

So,

le. `transform(students)`

returns

`[0, 1, 2]`

Now it converts text into numbers

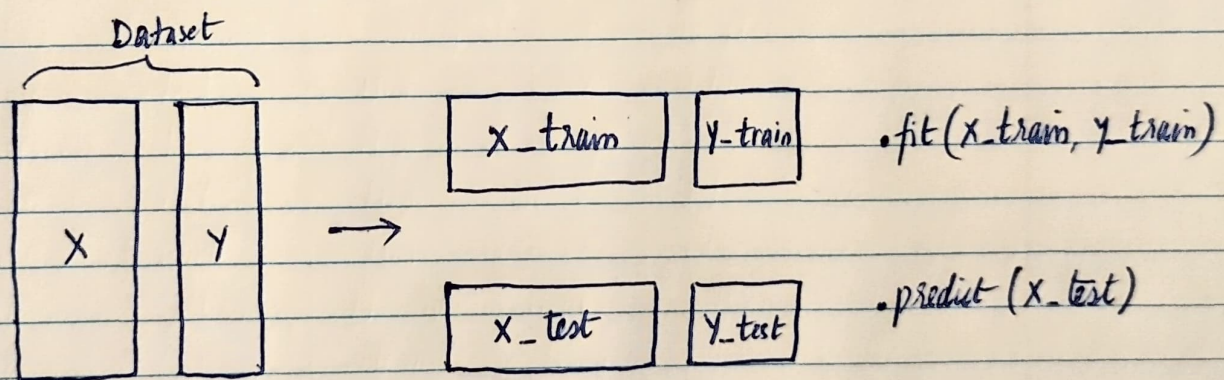
• `fit_transform()` ↘

it is just the combination of both `.fit()` &

`.transform()`

directly returns

`[0, 1, 2]`



Decision tree usually overfits and overconfident